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INTERNATIONAL QUALIFICATIONS

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International AS and A-level

Chemistry (9620)
CH05 Practical and synoptic
Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH05 JUNE 2024

Overall, students on this component scored very slightly better than in January and last June. Although some very good answers were seen from many students, a few seemed to have little experience of doing the required experiments. Sometimes, general, good practice ideas were applied to their answers that were irrelevant in that context.

QUESTION 01

1.1 This was mostly done well. A few lost a mark for only stating the colour.

1.2 Some clearly knew the equation but many did very badly. Many of these started with the incorrect formula and the released electron was not shown. A wide range of products was seen.

1.3 Many good, clear answers. Some mistakenly incorporated the carbonate ion into a complex ion. Na_2CO_3 on the left hand side was perfectly alright, but then the two Na^+ ions were needed on the right hand side.

1.4 Many correct answers but quite a few slips.

QUESTION 02

2.1 Most chose a method using a gas syringe and some made a reasonable attempt to draw it. There were some pretty strange designs but a labelled gas syringe with some kind of delivery tube gained the first mark. A proper cross-section through the apparatus with seals and gaps in the right places was needed for the second mark.

2.2 The source of error needed to be something that a student would not reasonably be able to control. There is a definite, unavoidable minimum time lag between starting the reaction and getting the apparatus sealed. It is almost impossible to read the time and the volume at exactly the same time. Leaky or otherwise faulty apparatus was not sufficient. One exception to this was that a sticky syringe might cause a problem, even though when checked before the experiment it had moved smoothly.

2.3 Some carried out the whole of this item very well. Some didn't draw the dots, some drew an unacceptable curve, some didn't show their tangent. Some just used the data in the table which meant (usually) that their stated gradient was too steep.

2.4 This was done quite well. Some misread their graph (even though the two final points were drawn for them). Most carried out the conversions and the rearrangement. Many forgot to convert their final answer into cm^3

2.5 There were many full, clear answers. Some didn't mention the per unit volume (or equivalent) part or for the more collisions part, per unit time.

2.6 The use of a water-bath was well known. The very slow reaction was a common, good answer. The idea of the hydrogen peroxide possibly freezing was accepted but not the water.

QUESTION 03

3.1 Most used sulfuric acid (with a few using hydrochloric acid instead (allowed)). Nitric acid would be dangerous and ethanoic acid would form an ester, so these were not accepted.

3.2 Arrows or words were acceptable. A few used both and this could cause a contradiction. If the arrows were clearly intended as pointers, benefit of doubt was given if the words were correct.

3.3 Most referred to the size of the bubbles. A few mentioned splashing but didn't include the idea of oxidising mixture being splashed across into the receiver. A few referred, incorrectly, to slowing the reaction down.

3.4 Most had the correct idea that ethanal would boil and ethanol and water would not. Care is needed when writing ethanal close to ethanol. Sometimes the A/O letters were hard to distinguish. Students would be well-advised to take a little extra time to make sure that there is no doubt.

A very small number contested that this was not a good temperature because there would be substantial evaporation of the ethanol because it was so close to its boiling point. This was given credit.

3.5 Many good answers were seen here too. Some got confused, perhaps by language, by suggesting that the ethanal should be distilled before it can evaporate.

3.6 Many gave good, full and clear answers. Many gave parts of the answer eg stating that ethanal did not have hydrogen bonding without stating what it DOES have. Many missed out the important phrase about between the molecules (or intermolecular).

3.7 There were many good answers. Some common mistakes were calculating the mass of ethanal and simply dividing it by the mass of ethanol, using the density data incorrectly and miscalculating the M_r of one or both of the substances.

3.8 Most gave a good answer but a few contradicted themselves eg use reflux and remove the condenser or have the condenser vertical and fit a lid.

QUESTION 04

This was an easy question.

QUESTION 05

This was an easy question.

QUESTION 06

This was an easy question.

QUESTION 07

This was a moderately challenging question. Options B and D were common distractors.

QUESTION 08

This was an easy question.

QUESTION 09

This was an easy question. Option B was a popular distractor.

QUESTION 10

This was an easy question. Although most correctly chose C, D was a significant distractor.

QUESTION 11

This was an easy question. B was a popular incorrect alternative.

QUESTION 12

This was an easy question.

QUESTION 13

This was an easy question.

QUESTION 14

This was an easy question. Incorrect answers were spread evenly across A and C.

QUESTION 15

This was a moderately challenging question. B was by far the most common error.

QUESTION 16

This was an easy question. B was the most popular distractor.

QUESTION 17

This was a moderately challenging question.

QUESTION 18

This was an easy question.

QUESTION 19

This was an easy question.

QUESTION 20

This was an easy question.

QUESTION 21

This was a moderately challenging question. D was a popular, but incorrect, choice

QUESTION 22

This was an easy question.

QUESTION 24

This was an easy question.

QUESTION 25

This was an easy question.

QUESTION 26

This was a moderately challenging question. B and D were roughly equally popular distractors.

QUESTION 27

This was an easy question, although C drew a lot of attention.

QUESTION 28

This was an easy question.

QUESTION 29

This was a moderately challenging question. C was a very popular incorrect choice.

QUESTION 30

This was an easy question.

QUESTION 31

This was a moderately challenging question. D was the most popular incorrect answer.

QUESTION 32

This was a moderately challenging question. Option B was almost as popular as C, which was the correct answer.

QUESTION 33

This was an easy question. A significant number incorrectly chose C.